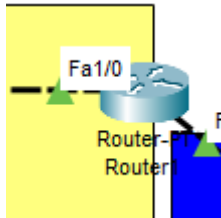


Router 2



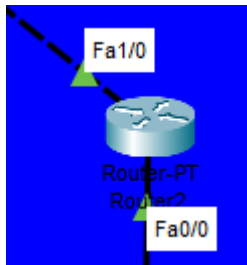
```
Router>show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

    10.0.0.0/30 is subnetted, 3 subnets
O E2   10.0.0.0 [110/20] via 10.0.1.1, 00:20:49, FastEthernet1/0
C       10.0.1.0 is directly connected, FastEthernet1/0
C       10.0.2.0 is directly connected, FastEthernet0/0
O E2   192.168.85.0/24 [110/20] via 10.0.1.1, 00:53:40, FastEthernet1/0
O E2   192.168.86.0/24 [110/20] via 10.0.1.1, 00:20:35, FastEthernet1/0
D      192.168.87.0/24 [90/30720] via 10.0.2.2, 00:50:57, FastEthernet0/0
D      192.168.88.0/24 [90/30720] via 10.0.2.2, 00:50:57, FastEthernet0/0

Router>
```

Router 3



```
Router>sh ip rou
Router>sh ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

    10.0.0.0/30 is subnetted, 3 subnets
D EX   10.0.0.0 [170/284160] via 10.0.2.1, 00:21:11, FastEthernet1/0
D EX   10.0.1.0 [170/284160] via 10.0.2.1, 00:51:19, FastEthernet1/0
C       10.0.2.0 is directly connected, FastEthernet1/0
D EX   192.168.85.0/24 [170/284160] via 10.0.2.1, 00:51:19, FastEthernet1/0
D EX   192.168.86.0/24 [170/284160] via 10.0.2.1, 00:20:57, FastEthernet1/0
C      192.168.87.0/24 is directly connected, FastEthernet0/0.30
C      192.168.88.0/24 is directly connected, FastEthernet0/0.40

Router>
```

PING

PC 1 a PC 2

```
Command Prompt

Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.86.10

Pinging 192.168.86.10 with 32 bytes of data:

Reply from 192.168.86.10: bytes=32 time<1ms TTL=127
Reply from 192.168.86.10: bytes=32 time=11ms TTL=127
Reply from 192.168.86.10: bytes=32 time<1ms TTL=127
Reply from 192.168.86.10: bytes=32 time=4ms TTL=127

Ping statistics for 192.168.86.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 11ms, Average = 3ms

C:\>
```

PC 3 a PC 4

```
Cisco Packet Tracer PC Command Line 1.0
C:\>PING 192.168.88.10

Pinging 192.168.88.10 with 32 bytes of data:

Reply from 192.168.88.10: bytes=32 time<1ms TTL=127
Reply from 192.168.88.10: bytes=32 time=9ms TTL=127
Reply from 192.168.88.10: bytes=32 time<1ms TTL=127
Reply from 192.168.88.10: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.88.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 9ms, Average = 2ms

C:\>
```

PC A PC 3

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.87.10

Pinging 192.168.87.10 with 32 bytes of data:

Reply from 192.168.87.10: bytes=32 time=1ms TTL=124
Reply from 192.168.87.10: bytes=32 time<1ms TTL=124
Reply from 192.168.87.10: bytes=32 time<1ms TTL=124
Reply from 192.168.87.10: bytes=32 time<1ms TTL=124

Ping statistics for 192.168.87.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>|
```

Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
	Successful	PC0	PC1	ICMP		0.000	N	0	(edit)	(delete)
	Successful	PC2	PC3	ICMP		0.000	N	1	(edit)	(delete)
	Successful	PC1	PC2	ICMP		0.000	N	2	(edit)	(delete)

Reflexión

¿Qué aprendí con esta tarea y cómo enfrenté los errores de configuración?

Con esta tarea aprendí a implementar una red utilizando varios protocolos de enrutamiento al mismo tiempo, específicamente RIP, OSPF y EIGRP, además de configurar comunicación entre VLANs utilizando un switch capa 3 y el método Router-on-a-Stick.

¿Cómo aplicaré el conocimiento de enrutamiento multi-protocolo en mi vida profesional?

El conocimiento de enrutamiento multi-protocolo será de gran utilidad en mi vida profesional, especialmente si trabajo en el área de redes o infraestructura tecnológica. En entornos reales muchas organizaciones utilizan diferentes protocolos de enrutamiento dependiendo de la arquitectura de su red, por lo que comprender cómo configurar RIP, OSPF y EIGRP

Conclusión

Esta tarea permitió integrar varios conceptos importantes de redes, como la segmentación mediante VLANs, el enrutamiento entre VLANs, la implementación de protocolos dinámicos y la redistribución de rutas entre diferentes dominios de enrutamiento. A través de la configuración práctica de la topología fue posible comprender cómo cada dispositivo cumple una función específica dentro de la red y cómo todos trabajan en conjunto para lograr conectividad total.